INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

ICE MAKING MACHINE, CUBE AND FLAKE (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers stainless steel, air-cooled condenser type ice making machines intended for use in messrooms, galleys, medical spaces, and other areas where a continuous supply of ice is needed.

2. CLASSIFICATION

Type 1: Cube Ice
- Size 1: 200 pound capacity
- Size 2: 300 pound capacity
- Size 3: 400 pound capacity
- Size 4: 500 pound capacity
- Size 5: 600 pound capacity
- Size 6: 800 pound capacity
- Size 7: 900 pound capacity
- Size 8: 1200 pound capacity

Type 2: Flake Ice
- Size 1: 300 pound capacity
- Size 2: 400 pound capacity
- Size 3: 600 pound capacity
- Size 4: 700 pound capacity
- Size 5: 800 pound capacity
- Size 6: 900 pound capacity
- Size 7: 1100 pound capacity

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The ice making machine shall be a complete, automatically operating self-contained unit with an ice making section and a storage bin. The machine shall be provided with a means of harvesting and storing ice. The ice making machine shall be provided with a water circulating system; motor or motor drives; necessary piping, wiring, controls, and safety devices; and any other components or accessories necessary to provide automatic operation and control of the ice level in the storage bin. The cabinet shall be provided with four removable legs, skids, or other means suitable for providing an 8-inch clearance between the underside of the ice bin and the finished deck, and for securing the machine directly to the deck. The machine shall have a depth of not greater than 25-3/4 inches and a height of not greater than 72 inches without legs.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 4110

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3.1.1 Ice making section. The ice making section (all components except the ice storage bin) shall be totally enclosed in a structurally sound housing with removable access panels for repair, replacement, and lubrication of internal components. Access panels shall be capable of detachment and re-attachment without special tools.

3.1.2 Storage bin. The storage bin shall be an integral part of the ice making machine. The thickness of the storage bin liner shall be not less than 24-gauge stainless steel. Insulation between the exterior and interior of the storage bin shall be of sufficient rating and density to prevent condensate from forming on the exterior surface of the storage bin when ice is stored in the bin. The insulation shall not be of the loose-fill type, shall not be combustible, and shall not release toxic fumes when exposed to a high temperature. The ice storage bin shall have a drainage opening. The ice storage bin shall be sized to contain the amount of ice produced by the ice making section during a 24-hour period. The bin cover shall be equipped with a key locking device.

3.2 Capacity. The ice making machine shall have the capacity to make the amount of ice for its size as listed in section 2, within a 24-hour period, under the following conditions: ambient air temperature not greater than 90 °F, and potable water temperature not greater than 70 °F. The ice making machine shall be capable of making and harvesting ice with a cooling value of not less than 122 BTU per pound.

3.2.1 Type 1. Ice produced by type 1 machines shall be cube ice. Cube ice normally refers to a fairly uniform product that is hard, solid, usually clear, and weighs less than 2 ounces per piece, as distinguished from flaked, crushed, or fragmented ice.

3.2.2 Type 2. Ice produced by type 2 machines shall be flake ice. Flake ice normally refers to a product that is frozen in thin sheets approximately 0.06 to 0.18 inch thick. The sheet may be flat or curved, and the thin ice is broken into random-sized flakes when harvested, as distinguished from cube, crushed, or fragmented ice.

3.3 Refrigeration unit. The refrigeration unit shall be an integral part of the complete ice making machine. The refrigeration unit shall be completely self-contained, pre-charged, and wired. The compressor and compressor motor shall be of the type and size to obtain the capacities listed in section 2. The motor shall be in accordance with NEMA MG-1.

3.3.1 Refrigerant (ozone depleting substance). The refrigerant shall be nonflammable, be in accordance with ARI 700, and be approved by the Environmental Protection Agency under the Significant New Alternate Program. It shall have an ozone depletion potential of not less than 0.05. (Shipboard requirement, see 7.2)

3.4 Electrical requirements. The unit shall operate on 115 VAC, 60 Hz, single-phase power, as defined in DOD-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. Mercury-type float switches shall not be used. (Shipboard requirement, see 7.2)

3.5 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of ARI 700, ARI 810, NFPA 70, NSF 12, and UL 563.

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3.6 Inclined operation. The unit shall operate satisfactorily, in accordance with the requirements of this CID (such as door remaining closed to prevent loss of contents) with no spillage of liquid or product when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 7.2)

3.7 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

3.8 Physical size limitations. The unit shall pass or shall disassemble so that it can pass through a shipboard hatch measuring 26 x 66 inches with 8-inch radius corners or a submarine access measuring 30 inches in diameter. (Shipboard requirement, see 7.2)

3.9 Accessibility. The unit shall be accessible from the front for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.10 Finish. The unit shall have the manufacturer's standard commercial exterior finish in accordance with NSF standards.

3.11 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.12 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.12.1 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

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7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Type and size (capacity) required
- When required, manuals shall be in accordance with ASTM F760

7.2 Shipboard requirement. Whenever a " (Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

7.3 Sources of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

DOD-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment

7.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A167 - Standard Specification for Stainless and Heat-resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM F760 - Food Service Equipment Manuals

7.3.3 American Refrigeration Institute (ARI) Standards. ARI Standards are available from the Air-Conditioning & Refrigeration Institute, 4301 North Fairfax Drive, Suite 425, Arlington, VA 22203.

ARI 700 - Specifications for Fluorocarbons and Other Refrigerants
ARI 810 - Automatic Commercial Ice-Makers

7.3.4 National Fire Protection Association (NFPA) Standards. NFPA Standards are available from the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.

NFPA 70 - The National Electric Code

7.3.5 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 12 - Automatic Ice Making Equipment

7.3.6 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 563 - Ice Makers

7.3.7 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

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7.4 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

Ice-O-Matic
6350 Alder Drive
Houston, TX 77081-4404

Scotsman Ice Systems
775 Corporate Woods Parkway
Vernon Hills, IL 60061

Vivian of St. Louis, Inc.
1525 Ocello Drive
Fenton, MO 63026-2480

Wilshire Corporation
2401 N. Palmer Drive
Schaumburg, IL 60196

MILITARY INTERESTS:

Custodian:
Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

Preparing activity:
Navy - SH
(Project 4110-0529)

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